

OTARU, Japan

WMO: 474110

Lat: 43.1825N

Lon: 141.0170E

Elev: 86

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.9	17.1	4.9	7.1	17.0	7.5	8.1	19.3	20.8	29.7	18.7	27.4	8.8	250	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	83.1	72.3	80.2	70.8	77.7	69.4	74.2	80.7	72.7	78.3	71.2	76.1	6.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	119.0	77.7	70.7	113.6	76.7	69.2	107.6	75.4	37.9	80.9	36.5	78.3	35.2	76.2	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.0	15.5	13.6	DB	11.4	89.2	3.0	2.6	9.2	91.1	7.5	92.6	5.8	94.0	3.5	95.9	(4)
			WB	10.0	76.0	2.9	2.1	7.9	77.5	6.2	78.7	4.6	79.8	2.5	81.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	25.9	26.9	33.3	43.9	54.2	61.2	68.9	71.6	65.0	53.3	40.7	30.0	(6)
		DBStd	16.90	4.92	5.39	5.66	6.09	5.86	4.99	4.49	4.30	5.06	5.68	7.31	5.63	(7)
		HDD50	3080	746	646	518	202	25	0	0	0	32	289	622		(8)
		HDD65	6620	1211	1066	983	632	339	133	14	2	61	362	730	1087	(9)
		CDD50	2359	0	0	0	19	154	337	585	670	449	136	9	0	(10)
		CDD65	424	0	0	0	0	2	19	134	208	61	0	0	0	(11)
		CDH74	1624	0	0	0	2	30	65	432	893	202	0	0	0	(12)
		CDH80	278	0	0	0	0	3	8	76	163	28	0	0	0	(13)
(14)	Wind	WSAvg	6.0	7.4	7.2	7.0	6.3	5.4	4.4	4.0	4.3	5.2	6.2	6.9	7.4	(14)
(15)	Precipitation	PrecAvg	55.2	4.9	4.1	3.5	3.3	3.1	4.2	6.2	5.1	6.3	4.3	5.0	5.4	(15)
(16)		PrecMax	63.3	7.2	7.9	5.8	5.0	4.3	6.0	9.3	9.6	11.8	7.2	8.9	8.7	(16)
(17)		PrecMin	42.2	2.5	2.2	1.5	1.5	1.8	2.9	4.1	2.3	2.4	2.0	2.6	3.3	(17)
(18)		PrecStd	5.5	1.2	1.3	1.1	1.0	0.7	0.8	1.6	2.1	2.8	1.3	1.6	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	41.8	44.7	52.6	69.9	78.1	80.3	86.4	88.7	83.8	71.0	62.5	49.9		(19)
(20)		MCWB	36.9	38.9	43.7	51.6	58.7	65.1	74.1	74.8	72.5	59.9	52.8	44.7		(20)
(21)		DB	37.8	40.1	48.1	62.1	72.1	75.2	81.9	84.0	79.0	67.4	57.6	43.9		(21)
(22)		MCWB	33.9	35.3	40.8	50.2	57.4	63.9	72.2	73.3	69.1	58.3	50.3	39.5		(22)
(23)		DB	35.4	37.5	44.7	57.4	67.5	72.2	78.6	81.2	75.8	64.9	54.6	40.9		(23)
(24)		MCWB	32.0	33.5	38.6	47.4	55.4	62.6	70.1	72.1	67.1	57.0	48.3	37.0		(24)
(25)		DB	33.2	35.4	41.9	53.6	63.7	69.4	75.7	78.7	73.2	62.2	51.6	38.2		(25)
(26)		MCWB	29.9	31.5	36.6	45.4	54.0	61.6	68.9	70.9	65.5	54.9	46.2	34.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	37.4	39.4	44.7	55.1	62.1	68.0	75.6	76.3	73.7	62.6	54.5	45.4		(27)
(28)		MCDB	40.9	43.7	51.0	66.0	73.3	74.9	83.7	84.3	81.0	66.7	60.1	49.6		(28)
(29)		WB	34.4	35.9	41.7	51.0	59.0	65.7	73.3	74.8	71.3	60.2	51.2	40.2		(29)
(30)		MCDB	37.3	39.4	46.9	59.8	68.3	72.8	79.7	81.5	77.2	65.4	56.5	43.6		(30)
(31)		WB	32.3	33.8	39.3	48.6	57.0	63.9	71.7	73.6	69.0	58.0	48.8	37.3		(31)
(32)		MCDB	35.0	37.2	44.0	56.2	65.0	70.3	76.5	79.3	74.2	63.6	53.8	40.6		(32)
(33)		WB	30.2	31.8	37.0	46.2	55.2	62.4	70.1	72.2	66.7	56.0	46.5	34.6		(33)
(34)		MCDB	33.0	35.0	41.4	52.9	62.1	68.0	74.6	77.4	71.9	61.4	51.3	37.8		(34)
(35)	Mean Daily Temperature Range	MDBR	8.4	9.3	10.4	13.5	14.7	12.8	11.2	11.5	13.0	13.0	10.9	8.5		(35)
(36)		5% DB	11.7	13.2	15.4	20.7	22.2	17.8	15.5	15.0	15.3	16.0	15.9	12.9		(36)
(37)		MCWBR	9.8	11.0	10.8	11.9	11.3	9.1	7.6	7.5	9.0	10.2	11.7	10.4		(37)
(38)		5% WB	11.8	13.1	15.2	19.5	19.3	15.3	13.5	13.0	13.5	14.5	15.5	12.9		(38)
(39)		MCWBR	10.3	11.3	11.2	11.9	10.7	8.2	7.0	6.9	8.6	10.2	12.2	11.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.350	0.399	0.464	0.524	0.539	0.522	0.504	0.469	0.415	0.409	0.393	0.367		(40)
(41)		taud	2.025	1.890	1.799	1.788	1.852	1.981	2.101	2.218	2.317	2.182	2.121	2.022		(41)
(42)		Ebn at Noon	233	239	240	237	238	243	246	251	257	238	217	211		(42)
(43)		Edh at Noon	36	48	60	66	63	56	49	43	36	36	32	33		(43)
(44)	All-Sky Solar Radiation	RadAvg	367	600	1014	1426	1624	1585	1440	1394	1183	825	449	304		(44)
(45)		RadStd	28	57	91	146	149	154	160	109	112	50	49	24		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page